

EXPERIMENT 27 : PREDICTION OF CATEGORICAL DATA USING DECISION TREE ALGORITHM THROUGH WEKA

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'J48 -C 0.25 -M 2'. The test options are set to 'Use training set'. The classifier output shows the following summary:

```
==== Summary ====
Correctly Classified Instances 147      98 %
Incorrectly Classified Instances 3      2 %
Kappa statistic 0.97
Mean absolute error 0.0233
Root mean squared error 0.108
Relative absolute error 5.1482 %
Root relative squared error 22.9089 %
Total Number of Instances 150

==== Detailed Accuracy By Class ====
```

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.960	0.010	0.980	0.960	0.970	0.955	0.990	0.970	Iris-setosa
	0.960	0.020	0.961	0.980	0.970	0.955	0.990	0.969	Iris-versicolor
	0.960	0.010	0.980	0.960	0.970	0.955	0.990	0.970	Iris-virginica

The confusion matrix is also displayed:

```
==== Confusion Matrix ====
a b c <-- classified as
50 0 0 | a = Iris-setosa
0 49 1 | b = Iris-versicolor
0 2 48 | c = Iris-virginica
```

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A decision tree visualizer window is overlaid on the main window, showing the structure of the J48 decision tree. The root node is 'petalwidth'. The tree structure is as follows:

```
graph TD
    Root((petalwidth)) -- "<= 0.8" --> Node1[ ]
    Root -- "> 0.8" --> Node2((petalwidth))
    Node1 -- "Iris-setosa (50/0)" --> Leaf1[Iris-setosa (50/0)]
    Node2 -- "<= 1.7" --> Node3((petalwidth))
    Node2 -- "> 1.7" --> Leaf2[Iris-virginica (48/0/1)]
    Node3 -- "<= 4.8" --> Leaf3[Iris-versicolor (48/0/1)]
    Node3 -- "> 4.8" --> Node4((petalwidth))
    Node4 -- "<= 1.5" --> Leaf4[Iris-virginica (3/0)]
    Node4 -- "> 1.5" --> Leaf5[Iris-versicolor (3/0/1)]
```

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'Bagging with 100 iterations and base learner'. The test options are set to 'Use training set'. The classifier output shows the following summary:

```
==== Summary ====
Correctly Classified Instances 150      100 %
Incorrectly Classified Instances 0      0 %
Kappa statistic 1
Mean absolute error 0.0156
Root mean squared error 0.0628
Relative absolute error 3.52 %
Root relative squared error 13.3147 %
Total Number of Instances 150

==== Detailed Accuracy By Class ====
```

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Iris-setosa
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Iris-versicolor
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Iris-virginica

The confusion matrix is also displayed:

```
==== Confusion Matrix ====
a b c <-- classified as
50 0 0 | a = Iris-setosa
0 50 0 | b = Iris-versicolor
0 0 50 | c = Iris-virginica
```

